Fiscal Year:	FY 2026	**Task Last Updated:**	FY 08/01/2025
PI Name:	Lundblad, Nathan Ph.D.		
Project Title:	Quantum Dynamics of Ultracold Bubbles		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	04240-6018	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2022 Physical Sciences NNH22ZDA001N-FP E.6 Fundamental Physics
Start Date:	10/04/2024	**End Date:**	10/03/2029
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA HQ
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Vishveshwara, Smitha Ph.D. (University of Illinois Urbana-Champaign) Sbroscia, Matteo Ph.D. (NASA Jet Propulsion Laboratory) Aveline, David Ph.D. (NASA Jet Propulsion Laboratory)		
Grant/Contract No.:	80NSSC25K7002		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We propose a flight research program focusing on the physics of ultracold atomic bubbles, a state of matter whose creation crucially demands a microgravity environment. We seek to observe the effects of altered geometry, topology, and dimensionality on quantum gases in previously unexplored ways, and answer open questions about quantum fluid (vortex) and non-equilibrium dynamics while pushing the techniques of atomic physics to new realms of applicability. Given the success of foundational work performed in the SM2-era Cold Atom Laboratory (CAL), which demonstrated the feasibility of ultracold bubble creation, we seek to expand the study of these delicate and unique structures. We aim to study the nature of the Bose-Einstein condensed (BEC) phase in the bubble: specifically, how condensation can persist when it emerges as a vestige of an inflated version of an initial BEC, and also if created through other means from an initial ultracold (but non-condensed) bubble. Understanding the nature of adiabaticity in bubble inflation remains an open question, and the dynamics of rapidly inflated (non-adiabatic) bubbles are unexplored. Additionally, we wish to explore the role of dimensionality as the bubble wall is varied from being `thick' (3D) to `thin' (2D), with distinct behaviors of vortices (potentially related to so-called Berezinskii–Kosterlitz–Thouless / BKT physics) and collective modes. Finally, we wish to explore the capacity of inflation to act as refrigeration (hinted at in initial studies) to prepare millimeter-scale quantum-coherent bubble samples at nK-scale temperatures. We will use the ultracold atomic physics capabilities of the CAL instrument, complemented by theoretical and computational modeling, as well as continued experimental testbed development. The technique of radiofrequency (rf) dressing is the core technology here, enabled in the CAL design through its capacity for rf-driven evaporative cooling. Using tailored ramps of applied rf magnetic fields, we can deform the trap provided by the CAL facility into a bubble-like structure of diverse shape and character. Using both axes of imaging aboard CAL will provide information about the bubble sample's density and momentum distributions. The increasingly sophisticated array of rf and microwave control options will enable new ideas such as nested bubbles, evaporative cooling within a bubble, and precision spectroscopy of the bubble state. Tailored modulation of the atom-chip magnetic fields (or rapid evaporation or `quenching') can enable the creation and observation of quantum vortices. In our associated ground-based work, the use of computational modeling of bubble physics (particularly, simulation of the Gross-Pitaevskii equation in realistic traps) will provide insight and guidance, as will further development of an independent atom-chip testbed and collaboration with international drop tower microgravity facilities. The proposed work deals in questions of strong current relevance at an intersection of atomic, quantum, many-body, condensed matter physics. These experiments require ongoing freefall in order to provide the pristine bubble confining forces for the ultracold samples, free from the distortion of gravitational tilt, and thus ideal for the International Space Station (ISS)-based user facility that CAL has established. The repetition rate and flight heritage of CAL makes it a unique observatory for quantum bubble physics. The theoretical physics community has expressed wide and varied interest in the study of quantum bubbles; from pure fluid dynamics interest (and the fundamental nature of vortex flow on curved manifolds) to the physics of phase transitions in nonstandard topologies, to understanding the superfluid physics in shell-shaped layers in neutron stars and cosmological models of the early Universe. In sum, the proposed research should expand our understanding of fundamental quantum systems and states of matter while broadening the reach and impact of spaceborne microgravity research.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work, while focused on the fundamental physics of ultracold atoms and not directly connected to human life, has a similar impact to life on Earth as that of all fundamental physics; it broadens our understanding of the physical world and helps us further cement our collective picture of quantum mechanics as "the way the world works." It explores the ways in which Bose-Einstein condensates with unique topology and geometry can be made, and to what extent the gravity-well of terrestrial labs render certain investigations difficult or impossible. The observations made aboard CAL through this project are a clear demonstration that physical insight can sometimes require microgravity facilities to be fully developed, and that spaceborne atomic physics experiments can be valuable contributions to our collective scientific efforts.
Task Progress:	Accomplishments during the reporting period were limited by the technical difficulties experienced by the NASA Cold Atom Laboratory (CAL) facility: namely, the need for replacement of the SM3 unit and the intermittent operation of the temporary replacement SM1 unit. Activities focused on computational modeling of the ultracold bubble state anticipated to be created aboard CAL, the thermodynamics of the bubble inflation process, and the technical details of the radiofrequency magnetic field inhomogeneity of the CAL instrument, thought to be a limiting factor in some aspects of the sought-after physics. A modicum of data runs were taken using the temporary CAL apparatus SM1; while some rf-dressing bubble structures have been observed, the field inhomogeneities present in this first-generation apparatus (as well as limitations to bubble aspect ratio caused by CAL atom-chip malfunctions) have thus far prevented observing Bose-Einstein condensate physics in the bubble state. Development of a ground testbed apparatus similar to one at NASA's Jet Propulsion Laboratory (JPL) continued at Bates, with the establishment of new optical benches and computer control for a ColdQuanta cold atom setup. This ground testbed is intended to provide more rapid prototyping of CAL experimental runs. An independent computational model for the CAL apparatus was finalized in collaboration with scientists at the Institute for Quantum Optics (IQO/Hannover), which agreed in broad strokes and many details with a Bates-local model. Theoretical work was highlighted by the PhD completion of Dr. Brendan Rhyno at the University of Illinois at Urbana-Champaign (UIUC), who defended his thesis, "Explorations of Engineered Bosonic Systems in Ultracold Atoms, Oscillator Networks, Quantum Spin Chains, and Space," on November 19, 2024.
Bibliography Type:	Description: (Last Updated: 07/13/2026)

Abstracts for Journals and Proceedings	Rhyno B, Müller TEG, Garcion C, Eric Charron, Gerent J-B, Lundblad N, Vishveshwara S, Gaaloul N. "Modeling thermodynamic and dynamic properties of Bose-Einstein condensate bubbles in microgravity." 2025 German Physical Society (DPG) Spring Meeting of the Atomic, Molecular, Quantum Optics and Photonics Section (SAMOP), Bonn, Germany, March 9-14, 2025. Abstracts. 2025 German Physical Society (DPG) Spring Meeting of the Atomic, Molecular, Quantum Optics and Photonics Section (SAMOP), Bonn, Germany, March 9-14, 2025. , Mar-2025
Abstracts for Journals and Proceedings	Rhyno B, Vishveshwara S, Lundblad N, Gaaloul N, Gerent J-B. "Dynamic simulations of Bose-Einstein condensate bubbles in space." American Physical Society (APS) Global Physics Summit, Anaheim, CA, March 16-21, 2025. Abstracts. American Physical Society (APS) Global Physics Summit, Anaheim, CA, March 16-21, 2025. , Mar-2025
Dissertations and Theses	Rhyno B. "Explorations of engineered bosonic systems in ultracold atoms, oscillator networks, quantum spin chains, and space." Dissertation, University of Illinois, November 2024. , Nov-2024
Significant Media Coverage	Vishveshwara S, Vishveshwara CV. "Two revolutions: Einstein’s relativity and quantum physics: A dialogue between father and daughter. Dr. Nathan Lundblad and Cold Atom Lab mentioned in popularizing-science history/memoir. " Oxford Academic online edition, Oxford University Press, December 9, 2024, https://doi.org/10.1093/9780191818080.001.0001 , Dec-2024